Year 13 Further Maths - Complex numbers and trigonometry

Similar questions - Multiple-angle formulae and exact values

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Use de Moivre's theorem to show that $\cos 5\theta \equiv 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. **[5]**

(b) By setting $\theta = \frac{\pi}{10}$, show that $16c^4 - 20c^2 + 5 = 0$ where $c = \cos\frac{\pi}{10}$. **[3]**

(c) Hence show that $\cos\frac{\pi}{10} = \sqrt{\frac{5+\sqrt{5}}{8}}$, justifying your choice of root. **[3]**

2

(a) By expanding $\left(z - \frac{1}{z}\right)^3$ where $z = e^{i\theta}$, show that $4\sin^3\theta \equiv 3\sin\theta - \sin 3\theta$. **[5]**

(b) Hence find the exact value of $\int_0^{\pi/3} \sin^3\theta \, d\theta$. **[3]**

Question 1

(a) $\cos 5\theta = \mathrm{Re}(\cos\theta + \mathrm{i}\sin\theta)^5 = c^5 - 10c^3s^2 + 5cs^4$ where $c = \cos\theta$, $s = \sin\theta$. With $s^2 = 1 - c^2$: $c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 = c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 = 16c^5 - 20c^3 + 5c$

(b) At $\theta = \frac{\pi}{10}$, $\cos\frac{\pi}{2} = 0$, so $16c^5 - 20c^3 + 5c = 0$. Dividing by $c \neq 0$: $16c^4 - 20c^2 + 5 = 0$

(c) $c^2 = \dfrac{20 \pm \sqrt{400 - 320}}{32} = \dfrac{5 \pm \sqrt{5}}{8}$. $\cos\frac{\pi}{10}$ is close to 1 (the angle is 18°) so c^2 is the larger value $\dfrac{5+\sqrt{5}}{8} = 0.905$ (the smaller, 0.345, is $\cos^2\frac{3\pi}{10}$). $c > 0$, so $c = \sqrt{\dfrac{5+\sqrt{5}}{8}}$

Question 2

(a) $\left(z - \frac{1}{z}\right)^3 = z^3 - 3z + 3z^{-1} - z^{-3} = (z^3 - z^{-3}) - 3(z - z^{-1})$. Left side is $(2\mathrm{i}\sin\theta)^3 = -8\mathrm{i}\sin^3\theta$; right side is $2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$. So $-8\sin^3\theta = 2\sin 3\theta - 6\sin\theta$, i.e. $4\sin^3\theta = 3\sin\theta - \sin 3\theta$

(b) $\frac{1}{4}\displaystyle\int_0^{\pi/3} (3\sin\theta - \sin 3\theta)\,\mathrm{d}\theta = \frac{1}{4}\left[-3\cos\theta + \frac{1}{3}\cos 3\theta\right]_0^{\pi/3} = \frac{1}{4}\left[\left(-\frac{3}{2} - \frac{1}{3}\right) - \left(-3 + \frac{1}{3}\right)\right] = \frac{1}{4}\cdot\frac{5}{6} = \dfrac{5}{24}$